

Review

Mechanisms and Efficacy of Chinese Herbal Medicines in Benign Prostatic Hyperplasia*

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ABSTRACT Benign prostatic hyperplasia (BPH) is one of the most common diseases in elderly men, the incidence of which gradually increases with age and leads to lower urinary tract symptoms (LUTS), which seriously affects the quality of life of patients. Chinese herbal medicines (CHMs) are widely used for the treatment of BPH in China and some other countries. To explore the molecular mechanisms of CHMs for BPH, we conducted a review based on peer-reviewed English-language publications in PubMed and Web of Science databases from inception to December 31, 2023. This article primarily reviewed 32 papers on the use of CHMs and its active compounds in the treatment of BPH, covering animal and cell experiments, and identified relevant mechanisms of action. The results suggest that the mechanisms of action of CHMs in treating BPH may involve the regulation of sex hormones, downregulation of cell growth factors, anti-inflammatory and antioxidative effects, inhibition of cell proliferation, and promotion of apoptosis. CHMs also exhibit α -blocker-like effects, with the potential to relax urethral smooth muscle and alleviate LUTS. Additionally, we also reviewed 4 clinical trials and meta-analyses of CHMs for the treatment of BPH patients, which provided initial evidence of the safety and effectiveness of CHMs treatment. CHMs treatment for BPH shows advantages as a multi-component, multi-target, and multi-pathway therapy, which can mitigate the severity of the disease, improve LUTS, and may become a reliable treatment option in the future.

KEYWORDS benign prostatic hyperplasia, Chinese herbal medicine, mechanism, efficacy

Benign prostatic hyperplasia (BPH) is one of the most common urological disorders in older men and can cause lower urinary tract symptoms (LUTS), such as storage LUTS (leading to urinary obstruction, diminished flow, voiding strain, or prolonged voiding) or voiding LUTS (leading to frequent and urgent urination, nocturia, and reduced urine stream), it also can affect post micturition LUTS (post-urinary drip or incomplete emptying),⁽¹⁾ seriously affecting the health and quality of life (QOL) of elderly male patients.⁽²⁾ With the advent of an aging population, the incidence and prevalence of BPH and LUTS are rapidly increasing, and in recent years there has been an upward trend in the proportion of men under the age of 50 years who present with symptoms.⁽³⁾ The exact molecular mechanism of BPH is unclear, but it has been confirmed that excessive cell proliferation (both epithelial and mesenchymal cells), as well as restricted programmed cell death processes, can cause an increase in prostate volume (PV).⁽⁴⁾ In the etiology of BPH, various factors such as androgens, estrogens, mesenchymal and epithelial cell interactions, growth factors, and neurotransmitters may affect the development and progression of BPH, either individually or in combination.⁽⁵⁾ Since the

etiology of BPH has not been fully elucidated and the pathophysiological changes of BPH are complex, there is few single drug that has shown good efficacy in treatment for BPH. Therefore, there is an urgent need to find more effective and safe drug treatment options for a wide population.

Chinese herbal medicines (CHMs) are widely

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used for the treatment of BPH in China and some other Asian countries,⁽⁶⁾ botanical preparations are widely used in Europe and the United States,⁽⁷⁾ and there is a significant increase in the use of CHMs and botanical preparations throughout the world due to their natural and low side-effect.⁽⁸⁾ Clinical trials and experimental studies have shown that Chinese medicine (CM) has an important role in improving clinical symptoms, reducing PV, and improving QOL in patients with BPH.⁽⁸⁾ To explore the molecular mechanisms of CHMs for BPH, we conducted a review based on peer-reviewed English-language publications. Our focus was on the anti-inflammatory, antioxidant, effects on sex hormones, and impacts on cell proliferation, and apoptosis associated with CHMs and their extracts commonly used in BPH treatment.^(9,10)

Search Strategy and Publications Selection

In PubMed and Web of Science databases, a search for studies on the treatment of BPH with CHMs was conducted from inception to December 31, 2023. We utilized the search strategy provided in Appendix 1 to retrieve all records from both databases. The eligibility criteria and data extraction of literature are detailed in Appendix 2. After excluding duplicate studies, 307 out of 496 searched studies were included for evaluation, as depicted in the selection flowchart (Appendix 3). Subsequently, following a review of the titles, abstracts, and full texts of these 307 studies, 271 were excluded for not meeting inclusion criteria. Ultimately, 36 studies were included, with 32 focusing on the mechanisms of CHMs treatment for BPH, and the remaining 4 being clinical studies, as detailed in Appendixes 4 and 5.

Mechanisms of CHMs in Treating BPH

Sex Hormone Regulation of CHMs

The imbalance of estrogen, androgen, and the ratio of estrogen to androgen is considered a significant factor in the development of BPH.^(11,12) The development, differentiation, proliferation, and survival of the prostate are largely dependent on androgens. Compared to testosterone (T), dihydrotestosterone (DHT) exhibits a higher affinity for androgen receptor (AR). Intracellular T is converted to DHT through the action of 5- α -reductase (5-AR), a process crucial in the regulation of androgen activity. Additionally, the aromatase enzyme converts androgens, such as T, into estrogens, like estradiol.⁽¹³⁾ Estrogen is also present in prostate, particularly in elderly males, where

estrogen levels may increase.⁽¹⁴⁾ Although estrogen's influence on prostate growth is relatively minor, it plays a role to some extent in the development of BPH.⁽¹⁵⁾ Steroid receptor coactivator 1 (SRC1) is a pivotal transcriptional coactivator that interacts with nuclear hormone receptors in a ligand-dependent manner, regulating gene expression. When hormones like androgens bind to their respective receptors, SRC1 can associate with the receptor-hormone complex, thus facilitating the transcription of target genes.⁽¹⁶⁾

CHMs have demonstrated the potential to significantly improve the imbalances induced by BPH, including the elevation of DHT levels, the reduction of T levels, and the increased expression of SRC1, reflecting their potential in modulating androgen mechanisms in BPH treatment. *Coptis chinensis* Franch (Huanglian) has been commonly used for more than 2,000 years, its main active compound berberine inhibited testosterone propionate (TP)-induced changes in epithelial cell line RWPE-1 cells, including increased cell proliferation and the expressions of prostate-specific antigen (PSA), 5-AR, and AR, through the extracellular signal-regulated kinase pathway.⁽¹⁷⁾ Bakuchiol, a natural compound abundant in *Psoralea corylifolia* L (Buguzhi), regulated estrogen and androgen signaling by down-regulating stromal aromatase and up-regulating epithelial estrogen receptor beta (ER β), further inhibiting BPH development.⁽¹⁸⁾ *Centella asiatica* (L.) Urban (Jixuecao), traditionally used for wound healing and anti-inflammation, has its aqueous extract enhancing BPH inhibition by targeting the AR and phosphatidylinositol 3-kinase (PI3K)/protein kinase B (Akt) pathways, potentially making it a candidate for BPH treatment formulations.⁽¹⁹⁾

Androgen promotes the proliferation of prostate stromal cells and the differentiation of epithelial cells through AR. In contrast, regarding estrogen, ER α has a proliferative function, while ER β exerts inhibitory effects on proliferation, based on their specificity for ERs.⁽²⁰⁾ CHMs approaches involve downregulating AR and ER α while upregulating ER β as potential treatments for BPH. PSA is highly expressed in prostate diseases, including BPH and prostate cancer. It is considered one of the typical androgen-dependent genes regulated through AR.⁽²¹⁾ Stimulation by DHT significantly upregulates PSA expression in human prostate stromal cell line WPMY-1 and epithelial cell

line RWPE-1, while *Psoralea corylifolia* L extract treatments significantly downregulate the mRNA and protein expressions of PSA. This suggests that the inhibition of PSA expression in DHT-stimulated RWPE-1 and WPMY-1 cells is associated with transcriptional regulation mediated by AR.⁽⁹⁾

α -Blocker-Like Effects of CHMs

" α -Blocker-like effects" refer to effects similar to those of α -adrenergic receptor antagonists (α -blockers). This typically implies that a certain medication or treatment approach can produce effects resembling those of α -blockers, namely, reducing the smooth muscle tension in the prostate tissue, thus alleviating symptoms caused by BPH. These effects may include improving urinary flow, reducing urinary frequency, urgency, and other symptoms associated with BPH, as well as relieving other discomforts caused by this condition.⁽²²⁾ Some herbal remedies have been researched and utilized because they exhibit properties similar to those of α -blockers in relaxing the smooth muscles in urethra, but may not entail the same side effects. *Coptis chinensis* Franch and berberine was found to inhibit the prostate enlargement in estrogen-androgen induced-BPH rats and improve bladder voiding parameters in normal rats, as well as the contraction of detrusor strips under acetylcholine stimulation via Ca^{2+} and rho-associated protein kinase (ROCK) signaling, suggesting a possible dual effect similar to that of α -blockers and 5-AR inhibitors.⁽²³⁾ *Phellodendron amurense* (Huangbai) has the potential to alleviate prostatic contractility triggered by electrical nerve stimulation and direct muscle stimulation, suggesting its possible effectiveness in treating urological disorders caused by prostatic urethral obstruction such as in the case of BPH.⁽²⁴⁾

Anti-inflammation Effects of CHMs

Inflammation has long been recognized as a significant factor in the development and progression of BPH.^(25,26) Extensive research has demonstrated the pivotal role of inflammatory processes in BPH pathogenesis. Inducible nitric oxide synthase (iNOS), an enzyme activated during inflammation, plays a crucial role by generating nitric oxide (NO), a pivotal signaling molecule known to induce vascular dilation and inflammatory responses.⁽²⁷⁾ In the context of BPH, the activation of iNOS and subsequent NO production can lead to inflammation and hyperplasia within prostate tissue.⁽²⁸⁾ The expressions of inflammatory

mediators such as tumor necrosis factor- α (TNF- α), interleukin (IL)-1 β and IL-6 are increased in prostate tissue, resulting in localized inflammatory responses that induce proliferation of prostate cells, contributing to the overall enlargement of prostate.⁽²⁹⁾ Arachidonic acid (AA), a lipid, can be metabolized into pro-inflammatory mediators such as prostaglandins. In BPH, alterations in the metabolism of AA may lead to inflammation and hyperplasia in prostate tissue.⁽¹¹⁾ Cyclooxygenase-2 (COX-2) and 5-lipoxygenase (5-LOX) are enzymes associated with inflammation, as they convert AA into prostaglandins and various other metabolites, which can trigger inflammation. Elevated COX-2 and 5-LOX levels are associated with BPH development, indicating their potential role in causing the condition.⁽³⁰⁾

CHMs have been shown to reduce the expressions of iNOS, COX-2, and inflammatory cytokines, thereby alleviating the severity of BPH in rodent models. Flavonoids, the primary constituents of *Anemarrhenae Rhizoma* (Zhimu), exhibit anti-inflammatory and anti-tumor properties, inhibiting BPH-related inflammation by targeting AA metabolism and interfering with COX and LOX pathways.⁽³¹⁾ *Abacopteris penangiana* (Jixuelian) is commonly used for blood circulation, blood stasis, edema, and inflammation, its extracts total flavanol glycosides and acid hydrolysate may potentially protect against BPH by reducing inflammation via suppression of both PI3K/Akt pathway and nuclear factor-kappa B (NF- κ B) pathway.⁽³²⁾ Berberine suppressed androgen-dependent BPH development by targeting NF- κ B-mediated pro-inflammatory signaling.⁽³³⁾ *Curcuma longa* L (Jianghuang), an anti-inflammatory remedy in CHM, its lipophilic fraction (*Curcuma* oil) has been found to alleviate BPH by inhibiting NF- κ B activity through the downregulation of phosphorylated P65 expression.⁽³⁴⁾ *Saussurea costus* (Muxiang) also attenuated BPH progression by modulating the NF- κ B-p65 signaling pathway, thus offering a promising approach for BPH prevention and treatment.⁽³⁵⁾ In addition, *Asteris Radix et Rhizoma* (Ziwan) may potentially alleviate BPH by reducing DHT levels through AR mediation, and by exerting pro-apoptotic and anti-inflammatory effects, highlighting its potential as a phytotherapeutic agent for BPH accompanied by inflammation.⁽³⁶⁾

Antioxidative Effects of CHMs

The relationship between oxidative stress and BPH has been a subject of significant interest.⁽³⁷⁾

Oxidative stress is a state of imbalance in the cellular redox equilibrium, typically caused by the accumulation of free radicals and oxidants, and it can have important implications for the health of prostate tissues. Total antioxidant capacity (TAOC) is an indicator that measures the overall ability of an organism to resist oxidative stress. Enhancing TAOC can help combat oxidative stress related to BPH and maintain the stability of prostate tissues.⁽³⁸⁾ Several key antioxidant enzymes play crucial roles in counteracting oxidative stress, including glutathione peroxidase (GPx), superoxide dismutase (SOD), and catalase (CAT), which help eliminate harmful oxidative species like hydrogen peroxide within cells.⁽³⁹⁾ Glutathione (GSH) is among the most essential endogenous antioxidants within cells, working in concert with GPx to neutralize free radicals and oxidants. Glutathione reductase (GR) is a critical enzyme that helps maintain GSH in its reduced state.⁽⁴⁰⁾ However, under conditions of oxidative stress, GSH may become oxidized into its disulfide form, known as oxidized glutathione. This oxidation process can potentially lead to damage in prostate tissues. In individuals with BPH, the activities of GPx and CAT may become dysregulated, leading to an exacerbation of oxidative stress.⁽⁴¹⁾ Glutathione S-transferase (GST) is an enzyme family that assists in breaking down harmful oxidative byproducts, thereby mitigating the extent of oxidative stress.⁽⁴²⁾ In the context of BPH, the activity of GST may be pivotal in preserving the antioxidant balance within prostate tissues. Malondialdehyde (MDA) serves as one of the markers of oxidative stress and is a product of lipid peroxidation, reflecting the extent of oxidative damage to cell membranes.⁽⁴³⁾ Elevated levels of MDA in BPH suggest the presence of oxidative stress.

CHMs that commonly used in the treatment of BPH have antioxidant effects through upregulating the expressions of GPx, GSH, CAT, and SOD, and suppressing the elevation of MDA levels, thus help maintain the oxidative-reductive balance within prostate tissues.^(38,44) Inflammation and oxidative stress are intricately linked. Inflammation can lead to an increase in oxidative stress, which can trigger an inflammatory response. This interplay can result in abnormal growth and pathological change of prostate tissue.⁽⁴⁵⁾ Heme oxygenase-1 (HO-1) and NAD(P)H: quinone oxidoreductase (NQO) both play protective roles in regulating oxidative stress and inflammation within cells. BPH induction significantly downregulated

the expressions of antioxidant genes, including HO-1, NQO, SOD1, and GPx-1, while treatment with berberine significantly increased the levels of these antioxidant markers.⁽³³⁾ *Acorus gramineus* (Shichangpu) has been reported to have various pharmacological effects, including suppressing the growth of mast cells. The hot water extract of its root has potent anti-inflammatory and antioxidative effects that could alleviate BPH by countering TP-induced inflammation and oxidative stress through suppression of the NF- κ B pathway.⁽⁴⁶⁾

Downregulation of Cell Growth Factors of CHMs

Recent research has highlighted the potential involvement of various growth factors in the pathophysiological processes underlying BPH development.⁽⁴⁷⁾ These growth factors may play critical roles through various mechanisms in BPH pathogenesis, such as cell proliferation, angiogenesis, inflammation, and tissue repair.^(48,49) Overexpression of growth factors like epidermal growth factor (EGF), fibroblast growth factor, and vascular endothelial growth factor (VEGF) could potentially contribute to abnormal prostate tissue growth, thereby promoting the progression of BPH.⁽⁵⁰⁾ Additionally, transforming growth factor- β may exhibit a dual role in BPH pathophysiology, both promoting cell proliferation and potentially inhibiting inflammatory responses and reducing tissue damage.⁽⁵¹⁾ CHMs represents a promising therapeutic approach for BPH, aiming to modulate growth factor activity. *Prinsepia utilis* Royle (Qingciji) leaves are used in the treatment of BPH, its active fraction could significantly decrease the expression of VEGF in prostate tissue of BPH rats.⁽⁵²⁾ *Herba Swertiae Mileensis* (Qingyedan), used for prostatitis and BPH, contains swertiamarin as its main component, which reduces the overexpression of VEGF, EGF, and basic fibroblast growth factor (bFGF) induced by T in BPH rats, potentially contributing to its therapeutic effects.⁽³⁸⁾

Pro-apoptotic Effects of CHMs

The imbalance between cell apoptosis and proliferation may be one of the important causes of BPH.⁽⁵³⁾ The size of the prostate depends on the balance between cell proliferation and programmed death in gland. Excessive cell proliferation and reduction of programmed death can both lead to an increase in PV.⁽⁵⁴⁾ The expression levels of anti-apoptotic genes and proteins in BPH epithelial cells are significantly higher than those in normal prostate cells.⁽⁵⁵⁾ It is speculated that the epithelial

cells show strong anti-apoptotic properties in the process of BPH lesions, and CHMs can reverse this pathway to treat BPH. The process of cell apoptosis is regulated by the Bcl-2 protein family, which mainly includes anti-apoptotic proteins Bcl-2, Bcl-xL, and pro-apoptotic proteins Bax.⁽⁵⁶⁾ Caspase-3, one of the most important apoptosis performers in the Caspase family of proteins, plays a key role in mediating apoptosis, and is the main effector factor in the process of apoptosis. Caspase-3 is located downstream of the cascade reaction, which causes cell death by degrading the corresponding substrate within the cell, and Bcl-xL and Bcl-2 can prevent apoptosis by inhibiting the activation of Caspase-3.⁽⁵⁷⁾ In endogenous apoptosis, Bcl-xL can inhibit the apoptosis process by disrupting the formation of Bax homodimers.⁽⁵⁸⁾

Scutellaria baicalensis Georgi (Huangqin) inhibits prostate growth by rebalancing Bax and Bcl-2 and enhancing Survivin expression, an upstream regulator of mitochondria-dependent apoptosis.⁽⁵⁹⁾ *Herba Swertiae Mileensis* and swertiamarin reduced the ratio of Bcl-2/Bax, ameliorated T-induced prostatic hyperplasia and collagen deposition.⁽³⁸⁾ *Curcuma* oil enhances apoptosis by reducing the Bcl-2/Bax expression ratio and increasing the expression of activated caspases-8, -9, and -3 in prostatic tissues, reinforcing the therapeutic advantages of this natural compound for treating BPH.⁽³⁴⁾ *Abacopteris penangiana* rhizomes and its acid hydrolysate exert anti-BPH effects by enhancing apoptosis through the inhibition of prostatic Bcl-2 expression and the activation of Bax and tumor protein 53.⁽⁶⁰⁾ *Panax ginseng* (Renshen) treats BPH by upregulating dynamin-related protein 1 expression, which modulates mitochondrial dynamics, triggering the cellular apoptosis pathway as evidenced by increased Bax/Bcl-xL ratio and promotion of caspase-3 and poly ADP-ribose polymerase (PARP) cleavage.⁽⁶¹⁾

Anti-proliferative Effects of CHMs and Other Mechanisms

Proliferating cell nuclear antigen (PCNA) is an auxiliary protein in cellular processes, involved in cell division, DNA replication, and cell cycle control.⁽⁶²⁾ During the cell cycle, various proteins such as Cyclin E, Cyclin A, Cdk2, Cyclin D1, and Cdk4 form complexes, participating in the regulation of different phases of the cell cycle.⁽⁴⁾ Additionally, phosphorylated retinoblastoma protein (pRB) and E2F transcription factor 1 (E2F1) are closely

associated with cell cycle progression and cell division. *Cornus officinalis* Sieb. et Zucc. (Shanzhuyu) exhibited anti-proliferation effects by suppressing the expression of PCNA and modulating E2F1/pRB-dependent cell cycle markers in prostatic epithelial cells.⁽⁶³⁾ *Ponciri Fructus* (Zhishi) exhibits anti-proliferative and antioxidant effects, reducing PCNA expression and increasing antioxidant enzyme activities, suggesting its potential in preventing BPH development and progression.⁽⁶⁴⁾ In WPMY-1 cells, tanshinone II A induced cell cycle arrest in the G₂/M phase with decreased cyclin B1 expression, whereas in RWPE-1 cells, it led to cell cycle arrest in the G₀/G₁ phase with reduced cyclin D1 expression.⁽⁶⁵⁾ Matrix metalloproteinase-2 (MMP2), a collagenase, plays a role in degrading fibronectin and laminin within the basement membrane and is implicated in prostate pathological processes. Sinomenine hydrochloride, one of the main active ingredients of *Sinomenium acutum* Rehderett Wilson (Qingfengteng), downregulates MMP2 expression in TP-treated castrated mice's prostate tissue, possibly aiding BPH improvement.⁽⁶⁶⁾ Hypoxia-inducible factor 1 alpha (HIF-1 α) is a transcription factor that plays a pivotal role in responding to fluctuations in oxygen levels within cells. In the context of BPH, the activation of HIF-1 α in response to hypoxia leads to the modulation of downstream genes involved in various processes, including angiogenesis, cell proliferation, epithelial-to-mesenchymal transition (EMT), and remodeling of the extracellular matrix.⁽⁶⁷⁾ During EMT, there is a decrease in the expression of epithelial markers such as E-cadherin, and an increase in the abundance of mesenchymal markers, including alpha smooth muscle actin (α -SMA), vimentin and cytokeratin, in proliferative prostate tissue.⁽⁶⁸⁾ Additionally, twist, acting as a transcription factor, may play a role in cell transformation and migration in BPH.⁽⁶⁷⁾ Swertiamarin treatment led to reduced levels of HIF- α , Twist, vimentin, and α -SMA, while increasing the expression of E-cadherin when compared to the untreated BPH control.⁽³⁸⁾

Clinical Studies of CHMs in Treating BPH

Randomized Controlled Trials of CHMs in Treating BPH

Herbal Saxifrage Tablet (虎耳草片, HST) has potential efficacy in improving the QOL, PV, and maximum urinary flow rate (Qmax) of patients with BPH, but compared with Terazosin Hydrochloride Hytrin, HST has a poor effect in improving the International Prostate Symptom Score (IPSS).⁽⁶⁾ In another study,

after 24 weeks of treatment, the Kangquan Recipe (康泉方) group showed a significantly greater reduction in IPSS (from 16.9 ± 5.6 to 12.5 ± 4.6) and significantly higher levels of Qmax and average urine flow rate, compared to the Cernilton (a plant extract, typically collected from pollen gathered by bees) group.⁽⁶⁹⁾ In one study evaluating the treatment of BPH with Longbishu Capsule (癃闭舒胶囊, LBS),⁽²²⁾ the results showed that after 12 months of treatment, the scores of Qmax of patients were significantly higher than those at baseline, and the IPSS score and residual urine volume (PVR) of patients in the LBS alone group and the combination of Western medicine (WM) and LBS group was obviously lower than that in the WM group (Appendix 5).

Systematic Review and Meta-analysis of CHMs in Treating BPH

In a meta-analysis of Qianlie Shutong Capsules (前列舒通胶囊, QLST) for BPH, 18 studies with 1,802 patients were reviewed. The treatment group, received QLST alone or with WM, outperformed the control group, which received WM (α blockers and 5-AR inhibitors). Results indicated that the treatment group had a better clinical response rate and lower IPSS than the control group. After a 3-month course of treatment, the treatment group also showed improved Qmax, PVR, and PV compared to the control group.⁽⁷⁰⁾ A systematic review and meta-analysis of 31 randomized controlled trials (RCTs) assessed the effectiveness and safety of CHMs in BPH. The findings demonstrated that CHMs were more effective than WM in enhancing QOL and reducing PV. Adverse event rates for CHMs were similar to placebos but lower than WM. Nevertheless, due to the poor methodological quality of the literature and the limited number of trials included, the evidence supporting the efficacy of CHMs in BPH remains relatively weak.⁽⁷¹⁾

Discussion and Perspectives

BPH is a proliferative disorder of glandular and interstitial cells caused by a number of factors, primarily due to a disruption of the balance between cell proliferation and apoptosis, resulting in increased cell proliferation or decreased apoptosis.⁽⁷²⁾ LUTS associated with BPH are caused by urethral obstruction and pathophysiologic increases in prostatic smooth muscle tone. Hormonal alterations, chronic inflammation, and oxidative stress are thought to play important roles in the development of BPH. The production and accumulation of DHT in the prostate

promotes cell growth and induces cell proliferation. A high estrogen/androgen ratio promotes prostate mesenchymal cell proliferation, inhibits apoptosis, and upregulates ER expression. The interdependence between sex hormones and peptide growth factors has been found and may disrupt the balance between cell proliferation and apoptosis, leading to BPH.⁽⁷³⁾ T-induced chronic inflammation is often associated with BPH.⁽⁷⁴⁾ Local tissue hypoxia and inflammation can induce BPH. Prostate tissue hypoxia releases reactive oxygen species that promote angiogenesis and release VEGF, which acts on mesenchymal and epithelial cells.⁽⁷⁵⁾ The presence of chronic inflammation in the prostate also significantly upregulates bFGF expression, and the release of large amounts of inflammatory cytokines and growth factors can upregulate the expression of apoptosis inhibitory factors (e.g., Bcl-2), leading to massive proliferation of prostate cells and the disease develops.⁽⁷⁶⁾ The pathogenesis and pathological basis associated with the intricate network of inflammatory factors, growth factors, and sex hormones that regulate the site, timing, and extent of hyperplasia remain controversial.

In this study, we mainly reviewed the therapeutic mechanism of CHMs on BPH. The evidence comes from both animal and cellular researches. The data reviewed in this article indicates that the use of CHMs for BPH treatment is beneficial. The mechanisms involved in CHMs treatment for BPH encompass the regulation of sex hormone levels, mitigation of oxidative stress and inflammatory responses, inhibition of cell proliferation and the expression of growth factors, as well as promotion of apoptosis (Figure 1). CHMs containing multi-component, multi-target, and multi-pathway has greater advantages in the treatment of BPH, which is worth further study in the future.^(61,77) With the application of modern pharmacological research techniques to CHMs such as network pharmacology, transcriptomics, and metabolomics, the efficacy and mechanism of CHMs in the treatment of BPH have been gradually revealed, and systematic studies have been conducted from single herbs to compound formulations.⁽⁷⁸⁾ We have also reviewed some clinical studies and meta-analyses, which suggest that certain CHMs treatments have demonstrated promising efficacy in clinical trials. These treatments have shown improvements in BPH patients' QOL, PV, Qmax, and other relevant indicators. Additionally, when compared to WM, CHMs

has exhibited a lower incidence of adverse events (Appendix 5). Currently in the clinic, CHMs is effective for mild to moderate BPH with storage and voiding symptoms, while surgery remains the preferred option for severe cases resistant to drug therapy.^(70,71) Combining CM with WM appears promising but lacks sufficient evidence.⁽⁷⁹⁾ Mild to moderate BPH is treated with CHMs based on syndrome differentiation, while severe cases benefit from combined CHMs and WM treatment to enhance efficacy, reduce complications, and shorten hospital stays.⁽⁸⁰⁾ Continual optimization of synergistic treatment protocols is crucial for providing practical guidelines to clinicians.

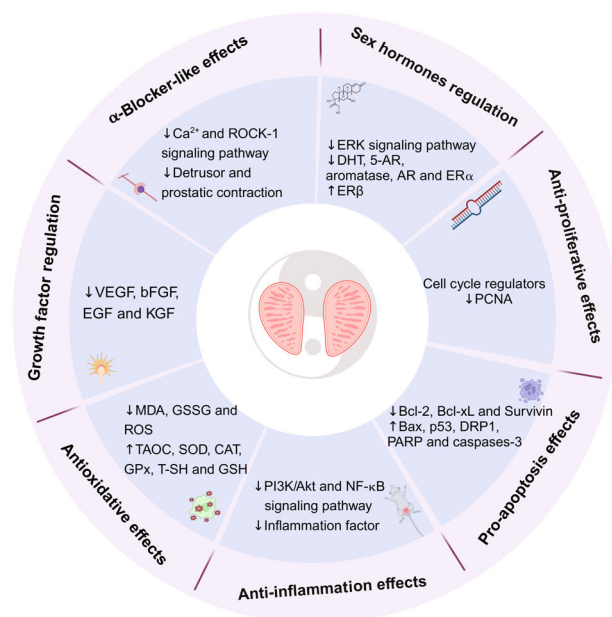


Figure 1. Mechanism of Chinese Herbal Medicines for BPH

Notes: ↓ : inhibited or alleviated; ↑ : upregulated or increased; ERK: extracellular signal-regulated kinase; DHT: dihydrotestosterone; AR: androgen receptor; ER: estrogen receptor; 5-AR: 5- α -reductase; VEGF: vascular endothelial growth factor; bFGF: basic fibroblast growth factor; EGF: epidermal growth factor; KGF: keratinocyte growth factor; NF- κ B: nuclear transcription factor- κ B; PI3K/Akt: phosphoinositide 3-kinase/protein kinase B; MDA: malondialdehyde; GSSG: glutathione disulfide; TAOC: total antioxidant capacity; SOD: superoxide dismutase; GSH: glutathione; CAT: catalase; T-SH: total sulfhydryl; GPx: glutathione peroxidase; PCNA: proliferating cell nuclear antigen; p53: tumor protein 53; PARP: poly ADP-ribose polymerase; DRP1: dynamin-related protein 1; ROS: reactive oxygen species; ROCK: rho-associated protein kinase

Regrettably, we did not provide detailed elaboration regarding animal and cellular models of BPH. Researchers should bear in mind the distinctive features, advantages, and drawbacks of various models when selecting the most suitable model for their specific research goals. This consideration is

of utmost importance for the effective planning of experimental designs and preclinical studies.⁽⁸¹⁾ In addition, there are also many challenges in the study of CHMs in the treatment of BPH. First, the prostate gland also contains other cell types, such as neurons and vascular endothelial cells, which may also play a role in prostate hyperplasia, and whether CHMs have any effect on them still needs to be studied in depth. Second, herbal prescriptions include multiple herbs, and the pharmacokinetic changes of these herbs in the human body need further study to elucidate their synergistic mechanisms of action. Third, there is a lack of multicenter, large-scale RCTs and long-term clinical studies based on evidence-based medicine to evaluate their efficacy. Future systematic evaluation of currently tapped CHMs for the treatment of BPH through pharmacokinetic and evidence-based medical studies is necessary.

Conclusions

CHMs show potential benefits in the treatment of BPH by modulating sex hormones, downregulating cell growth factors, reducing oxidative stress and inflammation, inhibiting cell proliferation, and promoting apoptosis. Modern pharmacological research techniques have revealed CHMs' efficacy and mechanism, emphasizing its multi-component, multi-target, and multi-pathway nature. Integrating CHMs with WM may offer a more effective and safer approach to BPH treatment, encouraging the medical community to move towards synergistic therapies and providing clinicians with practical guidelines.

Conflict of Interest

The authors declare that they have no competing interests.

Author Contributions

Responsible for the methodology and framework of this article: Wang F, Zhong CF and Li CL; data acquisition and management: Ma DY and Lyu DF; writing original manuscript: Wang F and Ma DY; editing and revision of articles: Li CL, Yang JT and Gao QH. All authors approve the submitted final version to be published.

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REFERENCES

1. Cornu JN, Oelke M, Parsons KF. Benign prostatic

- hyperplasia and lower urinary tract symptoms. *N Engl J Med* 2012;367:1668; author reply 1668-1669.
2. Chughtai B, Forde JC, Thomas DD, et al. Benign prostatic hyperplasia. *Nat Rev Dis Primers* 2016;2:16031.
 3. Qian S, Zhang S, Xia W, et al. Correlation of prostatic morphological parameters and clinical progression in aging Chinese men with benign prostatic hyperplasia: results from a cross-sectional study. *Prostate* 2021;81:478-486.
 4. Liu J, Liu D, Zhang X, et al. NELL2 modulates cell proliferation and apoptosis via ERK pathway in the development of benign prostatic hyperplasia. *Clin Sci (Lond)* 2021;135:1591-1608.
 5. El-Sherbiny M, El-Shafey M, El-Din El-Agawy MS, et al. Diacerein ameliorates testosterone-induced benign prostatic hyperplasia in rats: effect on oxidative stress, inflammation and apoptosis. *Int Immunopharmacol* 2021;100:108082.
 6. Li S, Lu A, Wang Y. Symptomatic comparison in efficacy on patients with benign prostatic hyperplasia treated with two therapeutic approaches. *Complement Ther Med* 2010;18:21-27.
 7. Moyad MA. Nutraceuticals and phytotherapy in men's health: antioxidants, pro-oxidants, and a novel opportunity for lifestyle changes. *Urol Clin North Am* 2022;49:239-248.
 8. Shen CY, Jiang JG, Yang L, et al. Anti-ageing active ingredients from herbs and nutraceuticals used in traditional Chinese medicine: pharmacological mechanisms and implications for drug discovery. *Br J Pharmacol* 2017;174:1395-1425.
 9. Kim HJ, Jin BR, An HJ. *Psoralea corylifolia* L. extract ameliorates benign prostatic hyperplasia by regulating prostate cell proliferation and apoptosis. *J Ethnopharmacol* 2021;273:113844.
 10. Zhang L, Fan XR, Xie H, et al. Anti-inflammatory and antioxidant effects of Kelong-capsule on testosterone-induced benign prostatic hyperplasia in rats. *Evid Based Complement Alternat Med* 2018;2018:5290514.
 11. Ho CK, Habib FK. Estrogen and androgen signaling in the pathogenesis of BPH. *Nat Rev Urol* 2011;8:29-41.
 12. Xue B, Wu S, Sharkey C, et al. Obesity-associated inflammation induces androgenic to estrogenic switch in the prostate gland. *Prostate Cancer Prostatic Dis* 2020;23:465-474.
 13. Nicholson TM, Sehgal PD, Drew SA, et al. Sex steroid receptor expression and localization in benign prostatic hyperplasia varies with tissue compartment. *Differentiation* 2013;85:140-149.
 14. Nicholson TM, Rieke WA. Androgens and estrogens in benign prostatic hyperplasia: past, present and future. *Differentiation* 2011;82:184-199.
 15. Asiedu B, Anang Y, Nyarko A, et al. The role of sex steroid hormones in benign prostatic hyperplasia. *Aging Male* 2017;20:17-22.
 16. Choi HM, Jung Y, Park J, et al. *Cinnamomi Cortex* (*Cinnamomum verum*) suppresses testosterone-induced benign prostatic hyperplasia by regulating 5 α -reductase. *Sci Rep* 2016;6:31906.
 17. Youn DH, Park J, Kim HL, et al. Berberine improves benign prostatic hyperplasia via suppression of 5 α reductase and extracellular signal-regulated kinase *in vivo* and *in vitro*. *Front Pharmacol* 2018;9:773.
 18. Miao L, Jiao C, Shao R, et al. Bakuchiol suppresses oestrogen/testosterone-induced benign prostatic hyperplasia development through up-regulation of epithelial estrogen receptor β and down-regulation of stromal aromatase. *Toxicol Appl Pharmacol* 2019;381:114637.
 19. Choi YJ, Fan M, Wedamulla NE, et al. Inhibitory effects of *Centella asiatica* (L.) Urban on enlarged prostate through androgen receptor and PI3K/Akt signaling pathways. *Food Funct* 2022;13:10235-10247.
 20. Royuela M, de Miguel MP, Bethencourt FR, et al. Estrogen receptors alpha and beta in the normal, hyperplastic and carcinomatous human prostate. *J Endocrinol* 2001;168:447-454.
 21. Sasaki T, Sugimura Y. The importance of time to prostate-specific antigen (PSA) nadir after primary androgen deprivation therapy in hormone-naïve prostate cancer patients. *J Clin Med* 2018;7:565-573.
 22. Song CS, Guo J, Chang DG, et al. Effect of Longbishu Capsule plus doxazosin on benign prostatic hyperplasia: a randomized controlled trial. *Chin J Integr Med* 2014;20:818-822.
 23. Miao L, Yun X, Yang X, et al. An inhibitory effect of berberine from herbal *Coptis chinensis* Franch on rat detrusor contraction in benign prostatic hyperplasia associated with lower urinary tract symptoms. *J Ethnopharmacol* 2021;268:113666.
 24. Xu Y, Ventura S. Extracts of bark from the traditional Chinese herb *Phellodendron amurense* inhibit contractility of the isolated rat prostate gland. *J Ethnopharmacol* 2010;127:196-199.
 25. Madersbacher S, Sampson N, Culig Z. Pathophysiology of benign prostatic hyperplasia and benign prostatic enlargement: a mini-review. *Gerontology* 2019;65:458-464.
 26. Tong Y, Zhou RY. Review of the roles and interaction of androgen and inflammation in benign prostatic hyperplasia. *Mediators Inflamm* 2020;2020:7958316.
 27. Li Y, Shi B, Dong F, et al. Effects of inflammatory responses, apoptosis, and STAT3/NF- κ B- and Nrf2-mediated oxidative stress on benign prostatic hyperplasia induced by a high-fat diet. *Aging (Albany NY)* 2019;11:5570-5578.
 28. Huang TR, Wang GC, Zhang HM, et al. Differential research of inflammatory and related mediators in BPH, histological prostatitis and PCa. *Andrologia* 2018;50:e12974.
 29. Ghadian A, Rezaei M. Combination therapy with omega-3 fatty acids plus tamsulocin and finasteride in the treatment

- of men with lower urinary tract symptoms (LUTS) and benign prostatic hyperplasia (BPH). *Inflammopharmacology* 2017;25:451-458.
30. Zhao X, Li Y, Huai J, et al. Compatibility effects of herb pair *Phellodendri chinensis* cortex and *Anemarrhenae rhizoma* on benign prostatic hyperplasia using targeted metabolomics. *Biomed Chromatogr* 2018;32:e4296.
 31. Cao X, Shang Y, Kong W, et al. Flavonoids derived from *Anemarrhenae Rhizoma* ameliorate inflammation of benign prostatic hyperplasia via modulating COX/LOX pathways. *J Ethnopharmacol* 2022;284:114740.
 32. Yang X, Yuan L, Xiong C, et al. *Abacopteris penangiana* exerts testosterone-induced benign prostatic hyperplasia protective effect through regulating inflammatory responses, reducing oxidative stress and anti-proliferative. *J Ethnopharmacol* 2014;157:105-113.
 33. Jin BR, An HJ. Oral administration of berberine represses macrophage activation-associated benign prostatic hyperplasia: a pivotal involvement of the NF- κ B. *Aging (Albany NY)* 2021;13:20016-20028.
 34. Wang S, Li Y, Li W, et al. *Curcuma* oil ameliorates benign prostatic hyperplasia through suppression of the nuclear factor-kappa B signaling pathway in rats. *J Ethnopharmacol* 2021;279:113703.
 35. Choi DH, Kim JY, An JH, et al. Effects of *Saussurea costus* on apoptosis imbalance and inflammation in benign prostatic hyperplasia. *J Ethnopharmacol* 2021;279:114349.
 36. Rho J, Seo CS, Park HS, et al. *Asteris Radix et Rhizoma* suppresses testosterone-induced benign prostatic hyperplasia in rats by regulating apoptosis and inflammation. *J Ethnopharmacol* 2020;255:112779.
 37. Nosalski R, Guzik TJ. Perivascular adipose tissue inflammation in vascular disease. *Br J Pharmacol* 2017;174:3496-3513.
 38. Wu X, Gu Y, Li L. The anti-hyperplasia, anti-oxidative and anti-inflammatory properties of Qing Ye Dan and swertiamarin in testosterone-induced benign prostatic hyperplasia in rats. *Toxicol Lett* 2017;265:9-16.
 39. Cipak Gasparovic A, Zarkovic N, Zarkovic K, et al. Biomarkers of oxidative and nitro-oxidative stress: conventional and novel approaches. *Br J Pharmacol* 2017;174:1771-1783.
 40. Sajjaboontawee N, Supasitthumrong T, Tunvirachaisakul C, et al. Lower thiol, glutathione, and glutathione peroxidase levels in prostate cancer: a meta-analysis study. *Aging Male* 2020;23:1533-1544.
 41. Ling S, Zhang H. Influences of chlorpyrifos on antioxidant enzyme activities of *Nilaparvata lugens*. *Ecotoxicol Environ Saf* 2013;98:187-190.
 42. Chen WZ, Pang B, Yang B, et al. Differential proteome analysis of conditioned medium of BPH-1 and LNCaP cells. *Chin Med J* 2011;124:3806-3809.
 43. Kucukdurmaz F, Efe E, Çelik A, et al. Evaluation of serum prolidase activity and oxidative stress markers in men with BPH and prostate cancer. *BMC Urol* 2017;17:116.
 44. Wang Z, Mao Q, Yuan Y, et al. Shuangshi Tonglin Capsule treats benign prostatic hyperplasia through the ROS/NLRP3 signaling pathway. *Int Urol Nephrol* 2024;56:1259-1271.
 45. Colado-Velázquez JI, Mailloux-Salinas P, Arias-Chávez DJ, et al. Lipidic extract of whole tomato reduces hyperplasia, oxidative stress and inflammation on testosterone-induced BPH in obese rats. *Int Urol Nephrol* 2023;55:529-539.
 46. Moon JM, Sung HM, Jung HJ, et al. *In vivo* evaluation of hot water extract of *Acorus gramineus* root against benign prostatic hyperplasia. *BMC Complement Altern Med* 2017;17:414.
 47. Devlin CM, Simms MS, Maitland NJ. Benign prostatic hyperplasia—what do we know? *BJU Int* 2021;127:389-399.
 48. Yang Y, Chisholm GD, Habib FK. Epidermal growth factor and transforming growth factor alpha concentrations in BPH and cancer of the prostate: their relationships with tissue androgen levels. *Br J Cancer* 1993;67:152-155.
 49. Wang Z, Zhang Y, Zhao C, et al. The miR-223-3p/MAP1B axis aggravates TGF- β -induced proliferation and migration of BPH-1 cells. *Cell Signal* 2021;84:110004.
 50. Rick FG, Schally AV, Block NL, et al. LHRH antagonist Cetrorelix reduces prostate size and gene expression of proinflammatory cytokines and growth factors in a rat model of benign prostatic hyperplasia. *Prostate* 2011;71:736-747.
 51. Li Y, Li J, Zhou L, et al. Aberrant activation of TGF- β /ROCK1 enhances stemness during prostatic stromal hyperplasia. *Cell Commun Signal* 2024;22:257.
 52. Peng Y, Peng C, Wu Y, et al. Chemical profiles of the active fraction from *Prinsepia utilis* Royle leaves and its anti-benign prostatic hyperplasia evaluation in animal models. *BMC Complement Med Ther* 2021;21:272.
 53. Gandour-Edwards R, Mack PC, Devere-White RW, et al. Abnormalities of apoptotic and cell cycle regulatory proteins in distinct histopathologic components of benign prostatic hyperplasia. *Prostate Cancer Prostatic Dis* 2004;7:321-326.
 54. Pagano E, Laudato M, Griffo M, et al. Phytotherapy of benign prostatic hyperplasia. A minireview. *Phytother Res* 2014;28:949-955.
 55. Minutoli L, Rinaldi M, Marini H, et al. Apoptotic pathways linked to endocrine system as potential therapeutic targets for benign prostatic hyperplasia. *Int J Mol Sci* 2016;17.
 56. Singh R, Letai A, Sarosiek K. Regulation of apoptosis in health and disease: the balancing act of BCL-2 family proteins. *Nat Rev Mol Cell Biol* 2019;20:175-193.
 57. Gabriel B, Sureau F, Casselyn M, et al. Retroactive pathway involving mitochondria in electroloaded cytochrome C-induced apoptosis. Protective properties of Bcl-2 and Bcl-XL. *Exp Cell Res* 2003;289:195-210.

58. He L, Perkins GA, Poblentz AT, et al. Bcl-xL overexpression blocks bax-mediated mitochondrial contact site formation and apoptosis in rod photoreceptors of lead-exposed mice. *Proc Natl Acad Sci U S A* 2003;100:1022-1027.
59. Jin BR, Chung KS, Kim HJ, et al. Chinese Skullcap (*Scutellaria baicalensis* Georgi) inhibits inflammation and proliferation on benign prostatic hyperplasia in rats. *J Ethnopharmacol* 2019;235:481-488.
60. Wei H, Wu G, Shi D, et al. Total flavan glycoside from *Abacopteris penangiana* rhizomes and its acid hydrolysate: characterisation and anti-benign prostatic hyperplasia potential. *Food Chem* 2012;134:1959-1966.
61. Park JY, Park WY, Song G, et al. *Panax ginseng* C.A. Meyer alleviates benign prostatic hyperplasia while preventing finasteride-induced side effects. *Front Pharmacol* 2023;14:1039622.
62. Elfakharany WA, Safwat MM, Essawy AS. Possible protective and curative effects of selenium nanoparticles on testosterone-induced benign prostatic hyperplasia rat model. *Folia Morphol (Warsz)* 2022;81:942-955.
63. Jin BR, Cheon SY, Kim HJ, et al. Anti-proliferative effects of standardized *Cornus officinalis* on benign prostatic epithelial cells via the PCNA/E2F1-dependent cell cycle pathway. *Int J Mol Sci* 2020;21:9567.
64. Jeon WY, Kim OS, Seo CS, et al. Inhibitory effects of *Ponciri Fructus* on testosterone-induced benign prostatic hyperplasia in rats. *BMC Complement Altern Med* 2017;17:384.
65. Wang C, Du X, Yang R, et al. The prevention and treatment effects of tanshinone II A on oestrogen/androgen-induced benign prostatic hyperplasia in rats. *J Steroid Biochem Mol Biol* 2015;145:28-37.
66. Fan MS, Xia YF, Ye RH, et al. Sinomenine hydrochloride can ameliorate benign prostatic hyperplasia by lowering the 5 α -reductase 2 level and regulating the balance between the proliferation and apoptosis of cells. *Molecules* 2023;28:803.
67. Kim HJ, Park JW, Cho YS, et al. Pathogenic role of HIF-1 α in prostate hyperplasia in the presence of chronic inflammation. *Biochim Biophys Acta* 2013;1832:183-194.
68. Wei P, Lin D, Zhang M, et al. Cryptotanshinone modulates proliferation, apoptosis, and fibrosis through inhibiting AR and EGFR/STAT3 axis to ameliorate benign prostatic hyperplasia progression. *Eur J Pharmacol* 2023;938:175434.
69. Huang YP, Wen YH, Wu GH, et al. Clinical study on Kangquan Recipe for benign prostatic hyperplasia patients: a randomized controlled trial. *Chin J Integr Med* 2014;20:949-954.
70. Liu SJ, Wang F, Gao QH, et al. Effect and safety of the Chinese medicine Qianlie Shutong Capsules in the treatment of benign prostatic hyperplasia: a meta-analysis. *Natl J Androl (Chin)* 2019;25:1021-1030.
71. Ma CH, Lin WL, Lui SL, et al. Efficacy and safety of Chinese herbal medicine for benign prostatic hyperplasia: systematic review of randomized controlled trials. *Asian J Androl* 2013;15:471-482.
72. Huang X, Lee C. Regulation of stromal proliferation, growth arrest, differentiation and apoptosis in benign prostatic hyperplasia by TGF-beta. *Front Biosci* 2003;8:s740-749.
73. La Vignera S, Condorelli RA, Russo GI, et al. Endocrine control of benign prostatic hyperplasia. *Andrology* 2016;4:404-411.
74. Anim JT, Udo C, John B. Characterisation of inflammatory cells in benign prostatic hyperplasia. *Acta Histochem* 1998;100:439-449.
75. Wang L, Yang JR, Yang LY, et al. Chronic inflammation in benign prostatic hyperplasia: implications for therapy. *Med Hypotheses* 2008;70:1021-1023.
76. Song W, Li DY, Yuan HC, et al. Relationship between interleukin-8 levels in expressed prostatic secretion and expressions of bFGF and Bcl-2 in benign prostatic hyperplasia. *Natl Med J China (Chin)* 2016;96:104-107.
77. Csikós E, Horváth A, Ács K, et al. Treatment of benign prostatic hyperplasia by natural drugs. *Molecules* 2021;26:7141.
78. Tao R, Liu E, Zhao X, et al. Combination of *Ligustri Lucidi Fructus* with *Ecliptae Herba* and their phytoestrogen or phytoandrogen like active pharmaceutical ingredients alleviate oestrogen/testosterone-induced benign prostatic hyperplasia through regulating steroid 5- α -reductase. *Phytomedicine* 2022;102:154169.
79. Song CS. Thinking and methods for treatment of benign prostatic hyperplasia with integrative medicine. *Chin J Integr Tradit West Med (Chin)* 2008;28:747-749.
80. Xu B, Liu M, Liu Y, et al. Risk factors of urinary pathogenic bacteria infection after benign prostatic hyperplasia surgery and curative effect analysis of Shuangdong Capsule intervention. *Emerg Med Int* 2022;2022:4069787.
81. Zhang J, Zhang M, Tang J, et al. Animal models of benign prostatic hyperplasia. *Prostate Cancer Prostatic Dis* 2021;24:49-57.

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